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Computational Studies of the Spatial Architecture of Primate Visual Cortex

Columns, Maps, and Protomaps

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1. Introduction

Computational neuroscience is a term that has recently come into widespread use, following a symposium* which was organized in the mid-1980s for the purpose of defining an area of research that had more contact with the biology of the brain than was (and is) customary in areas such as neural networks, but which involved significant mathematical and computational techniques and ideas. Some of the themes discussed in this symposium [later published in book form (Schwartz, 1990)] were that the study of brain form and function necessarily involved a hierarchy of spatial scales, and that computational techniques were associated with neuroscience in two complementary ways:

1. The application of methods of computer graphics, image processing, and numerical analysis is critical to the description of the nervous system.

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2. Understanding of the possible functional utility of neural architectures is likely to feed back to computation via the design of machine vision, robotics, and other applications.

In the present chapter, a review of the spatial architecture of primate V1 will be provided, which illustrates the issue of multiple scales of analysis, the descriptive use of computational method, as well as the use of neural architectures to motivate the construction of machine vision systems. These illustrations will be taken from a range of scale that might be termed *supraneuronal.* Cortex is viewed in terms of spatial patterns in which the scale of individual neurons is not addressed. This is much like an approach to fluid mechanics in which the individual molecules are ignored, in favor of large-scale variations in density and velocity fields. Many of the mathematical concepts which are central to classical fluid mechanics, such as the representation of fluid flows in terms of conformal mapping and patterns of vorticity, appear directly in a continuum level approach to cortical patterns, as will be made clear below. And, lest the reader become uneasy with a "continuum" approach to the nervous system,* it should be pointed out that this type of approach is implicit in many of the important anatomical techniques which have been developed during the past two decades. Thus, 2-deoxyglucose (2-DG), cytochrome oxidase (CO), and optical dye recording are ipso facto pattern level methods of description. What has been lacking in this area is not the application of pattern level description, but rather a careful mathematical analysis of the implications of those pattern level techniques which are the principal methods of characterizing the architecture of the cortex.

The first section of this chapter will review the columnar aspect of V1. There are three major columnar systems that have been described in striate cortex:

1. Ocular dominance "columns," whose clearest structure in the primate occurs promptly at the input in layer IV, and whose pattern is reminiscent of "zebra stripes" and other morphological patterns
2. Orientation columns whose somewhat regular pattern is pierced by a net of the more recently discovered CO puffs
3. Cytochrome oxidase puffs (Hubel and Livingstone, 1985), which consist of neurons with receptive fields that are chromatically tuned but orientationally nonspecific.

The spatial structure of both the ocular dominance and orientation column systems has been the subject of many computational models during the past 15 years. In this chapter, we show that the patterns of both systems, and most of the models which have been advanced to account for them, are underlaid by a simple model based on bandpass-filtered random (noise) patterns (Rojer and Schwartz, 1990a). Moreover, the orientation column–puff system may be viewed as a single system whose spatial pattern is characterized jointly by filtered noise, and the

*Most workers trained as experimental neuroscientists appear to have a strong preference for computational models based on reducing the conceptual units of the nervous system to their smallest physical scale, i.e., the level of synapses if not macromolecules. However, in the areas of interest of this chapter, it is not clear that synaptic level approaches have provided much insight into the pattern level phenomena addressed in this paper, and it might be argued that, in principle, they are not likely to do so.